

Joshua Himmens

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Engineering Physics Student

Technical Skills

Machine Learning | TensorFlow, Keras, PointNet, Weights and Biases (wandb), ONNX, LeRobot, Runpod, Triton

Embedded Programming | Experienced with FreeRTOS on TMS570, RP2040, STM32, Xilinx and Altera FPGAs, KiCAD

Software Development | Python, Rust, C, C++, Java, MATLAB, Bash, Git

Quantum Computing | Qiskit, QASM, Pasqal, QCNNS, VQCs

Particle Physics | ROOT, CERN grid computing, Athena

Work Experience

Tactile Sensing Engineering Intern, Optimus

Tesla | May 2025 – Dec 2025

- Designed tactile sensors using data-driven models to guide sensor architecture and manufacturing specifications.
- Collaborated on neural and classical physics models for Optimus tactile sensors.
- Characterized tactile sensors, defined test specification limits for incoming samples, and collaborated on test infrastructure design.

Associate Machine Learning Developer

AltaML | Jan 2025 – Apr 2025 | himmens.com/altaml

- Developed machine learning models using scikit-learn to save the Government of Alberta >\$4 million annually.
- Developed novel evaluation metrics to minimize regional bias and maximize transferability of models.
- Built Azure ML pipelines for automated model training and deployment, serving predictions via REST APIs integrated with SQL databases and Blob Storage.
- Implemented Uber's H3 hexagonal spatial indexing to balance geographic accuracy with user privacy in an infrastructure planning tool.

ATLAS Deep Learning Research Student

TRIUMF | May 2024 – Feb 2025 | himmens.com/triumf

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting-edge transfer learning approaches.
- Implemented models into production using NVIDIA Triton.

Command and Data Handling (CDH) Lead and Firmware Developer

UBC Orbit Satellite Design Team | Oct 2023 – Present | himmens.com/orbit

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of **10** firmware developers, with over **\$15 000** worth of technical infrastructure.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

Embedded Firmware Developer

UBC Orbit Satellite Design Team | Jan 2023 – Present | himmens.com/orbit

- Programmed device drivers, electrical ground support equipment (EGSE).
- Developed the ALEASAT Avionics Test Bench (FlatSat).
- Worked on the ALEASAT onboard computing system using FreeRTOS on a TMS570 microcontroller.

Publications and Presentations

Machine Learning for Robotic Napkin Folding at 2026 IEEE Vancouver Section AGM Mar 2026

- Placed 3rd. Explored various policies on SO-101 arms.

Developing Machine Learning Techniques for Particle Flow in the ATLAS Experiment at Canadian Astroparticle Physics Summer Student Talks Competition (CASST 2024) Jun 2024

- Placed 2nd of 44 presentations.

JetPointNet: A Machine Learning Approach to Cell-to-Track Attribution in the ATLAS Experiment at Canadian Undergraduate Physics Conference Oct 2024

3D Particle Flow in the ATLAS Calorimeter: How to Train Your Model at TRIUMF Science Week May 2024

ALEASAT ESA "Fly Your Satellite!" Training Week Presentation at European Space Agency's ESEC-GALAXIA (Transinne, Belgium) Mar 2024

Leadership and Advocacy

Chair

UBC IEEE Aerospace Electrical Systems Student Branch Chapter Aug 2025 - May 2026

- Served as *Technical Program Chair* for the Student Aerospace Summit, September 2025.
- Organized events to unify aerospace-interested students across UBC.

Advisory Team Member

Child Rights Connect Jan 2021 - Dec 2023

- Provided guidance to UN delegations on communication strategies for high-level rights goals.
- Presented to governments and consulted on international initiatives to support the UN Convention on the Rights of the Child.

Additional Experience

- "FizzVolt, HUD RL for Hardware Hackathon"** Built a GRPO reinforcement-learning environment that trains Qwen3 models to generate KiCad circuit designs using deterministic electrical-rule, cost, size, and availability validators as the reward
- "Quantum School for Young Students"** Participant in intensive quantum computing program
- "GENQ Global Hackathon (Geneva, 2025 and Calgary, 2024)"** Build QCNNS and QUBO solvers for quantum machine learning tasks

Honors and Awards

2026 Design & Innovation Day Faculty Award Apr 2026

Earned one of three Faculty Awards for Linique, a dual-arm napkin-folding robot powered by a four-camera learned controller and vision-language-action model; contributed improvements to the open-source LeRobot framework.

Hector J. MacLeod Award Apr 2024

Awarded to students who have demonstrated pioneering efforts in their chosen field of study and attained high scholastic honours.

Erich Vogt First Year Summer Research Experience (FYSRE) Award May 2024

Awarded to promising students in physics for a 4-month research placement.

Alberta Centennial Award and Alberta Premier's Citizenship Award May 2023

Awarded for outstanding community service.

Calgary Flames Foundation Community Involvement Scholarship Aug 2023

Awarded for community involvement.

Julia Turnbull Leadership Award for exceptional community service Jul 2023

Awarded for exceptional community service.